Homework 3

Reproducing plots and exploring real spatial data
Due: Nov 1st | Total: 75 points

Instructions

Submit **`HW3.R` or `HW3_report.pdf`**, together with a link to your GitHub repository. Put short explanations in the report or in `#` comments, and label answers by problem. Include the actual generated plots, not screenshots of the RStudio interface.

```
# Install packages once in the Console if needed:
# install.packages(c("ggplot2", "dplyr", "tidyr", "readr", "sf"))
library(ggplot2)
library(dplyr)
library(tidyr)
library(readr)
library(sf)

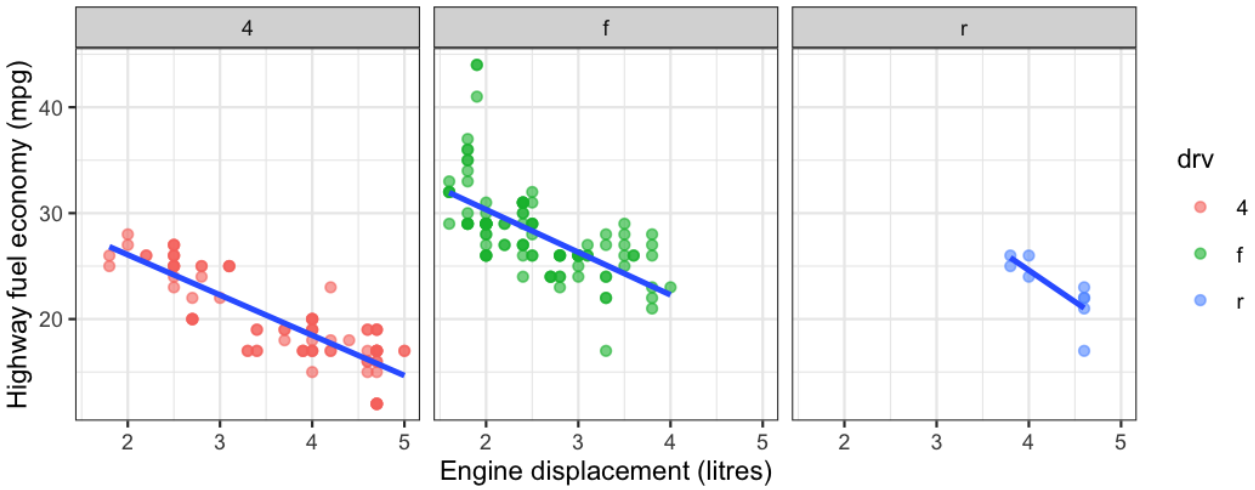
vehicles <- ggplot2::mpg       # Full dataset
gems <- ggplot2::diamonds     # Full dataset
```

Problem 1: Reproduce and explain plots (25 points)

Use the full package datasets defined on page 1. Reproduce the **data, mappings, layers, grouping, axis labels, and panels** in the figures. Use `theme_bw()` and readable legends. The calculations and graphical meaning must agree.

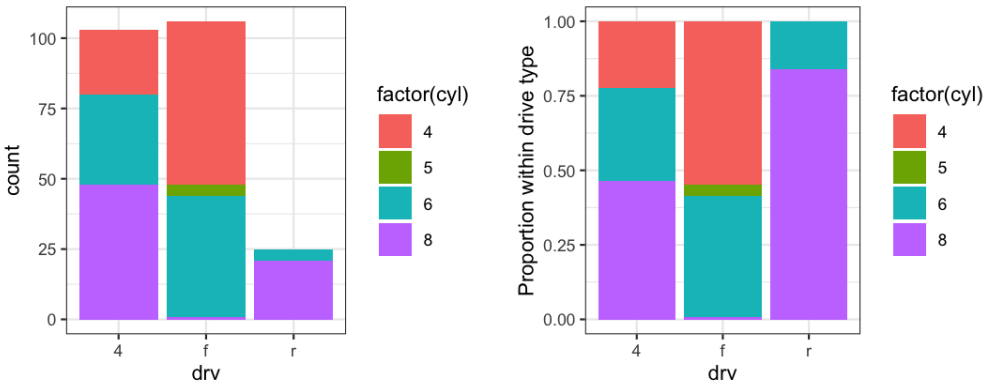
Part A (8 points)

Reference A. Engine size and highway fuel economy, separated by drive type.



- **(6 points)** Reproduce the same plot as above, using ggplot.
- **(2 points)** Explain why there are three fitted lines even when `drv` is mapped to color only in the point layer.

Part B (8 points)



- **(5 points)** Reproduce the same plot as above using ggplot.
- **(3 points)**

Now plot the proportion as y-axis. Submit the R code. Briefly explain why equal total heights in the right plot do not mean that the drive types have equal sample sizes.

Part C (9 points)

- **(5 points)** Create two count histograms of `carat` from the **same `smaller` table**. Save them as `p_fine` and `p_coarse`. Use bin widths `0.01` and `0.5`, respectively; in both cases use `boundary = 0` and `closed = "right"`. Keep the same x-axis range and labels, show both plots, and label each bin width. Report how many records were retained and excluded.
- **(2 points)** From the original `gems`, display the numbers of diamonds recorded as exactly `0.99` and `1.00` carat. Use code to answer.
- **(2 points)** In at most four sentences, describe a feature that is easier to see with the narrower bins.

```
# Your plot code, tables, and explanations for Parts A-C.
```

Instructions for Problems 2 and 3

Choose **one city or compact local region** and one kind of public facility, such as libraries, fire stations, or public charging sites.

Use **real, documented, downloadable data**, rather than the built-in example in R or fictional coordinates. Points will be deducted if you use the built-in data in R. A point CSV must include coordinates; a file containing only addresses is not sufficient. Suitable formats include CSV, GeoJSON, GeoPackage, or a complete zipped shapefile. Note that a map screenshot is not a spatial dataset.

Select a study area with **at least 20 usable facility records** and **at least 5 area units**. Use all eligible records in that area after your documented cleaning. If you use an API, check pagination so the download does not silently stop after its first page.

A suggested dataset, or find your own

The City of Chicago provides both of the following real datasets:

Needed data	Official source and download guidance
Library locations	Libraries: locations, contact information, and usual hours. Inspect the location field; it contains latitude and longitude.
Area boundaries	Boundaries: Community Areas.

Use the dataset's Export option and select a geographic format such as GeoJSON. |

You may use this pair or locate an equivalent pair through **Kaggle, Data.gov, or a government open-data portal**. For Kaggle, cite both the Kaggle page and the original publisher when available. Verify the license, coordinate information, and dates; public availability alone is not a claim that the records are current or complete.

Problem 2 (25 points)

Using real facility data for your chosen region, investigate: **Where are the facilities, and how much area lies within 500 meters or 1,000 meters of at least one facility?**

Instructions

Data. Do not remove facilities just because they share coordinates.

Maps. Include titles, clear layer labels or a legend, and source/date captions. You may display the result in a table or plot.

Areas. Use the same sites, count overlap only once, and include areas outside the study boundary. Check that the 1,000-meter area is at least as large as the 500-meter area.

Expected outputs

Submit your code and the **following outputs**:

Data. Record the source in `data_sources.csv` and keep the raw data unchanged. Provide cleaned locations with unique IDs and facility names. Report record counts before and after cleaning, coordinate problems, duplicates, and any removals with reasons. Save the points as `sites_m` in a suitable local projected CRS with meter units; state the CRS and units.

Maps. Create `p_sites` showing all retained facilities and `p_buffers` showing the facilities and their combined 500-meter buffer area.

Areas. Create `buffer_summary` with `radius_m` and `union_area_km2` for the 500-meter and 1,000-meter buffers. Keep the points and both combined buffer zones.

```
# Your code, data checks, two maps, area table, and interpretation.
```

Problem 3 (25 points)

Using the facilities from Problem 2 and a separate real boundary dataset, investigate: **Which areas have the most facilities, and does the comparison change when we account for area size?**

Instructions

Matches and summary. Note: You should count only sites matching exactly one area. Count each assigned site once and keep zero-count areas. Check that areas are positive and counts total the number of assigned sites.

Maps. Use the same boundaries, projection, map extent, but separate numeric legends with the correct units. Show zero-count areas and distinguish zero from missing values. Include titles and source/date captions stating how many sites were left unassigned.

Expected outputs

Submit your code and the following outputs:

Boundaries. Record the source in `data_sources.csv`. Provide valid polygons with unique IDs and area names, using the same projected CRS as `sites_m`.

Optional: Report missing or duplicate IDs, empty or invalid shapes, and any repairs or removals. Show a map with the sites over the boundaries, note any differences in geography or dates, and keep areas with no sites.

Matches and summary. Save each site's ID and number of area matches in `site_match_audit.csv`. Report how many sites match zero, one, or multiple areas, and identify sites without exactly one match.

Create `district_counts` with `district_id`, `district_name`, `n_sites`, `area_km2`, and `sites_per_km2`. Output the table of the result.

Maps. Create `p_count` for facility counts and `p_density` for facilities per square kilometer.

```
# Your code, boundary checks, matching audit, summary table,
# two comparison maps, and interpretation.
```